

[Affiliated College Name]
University of the Punjab, Lahore

Final Year Project

Final Documentation Format Guidelines

Template / Guidelines

Note: This template is intended as a general guide. Students may add or skip sections depending on the nature and needs of their project. Ensure that the document clearly reflects your system's requirements and design approach.

[Title of Final Project]

Team ID: [##]

Session: BSCS Fall [####]

Project Advisor: [Name of Project Advisor]

Submitted By:

Name of Student	Roll Number
[Student Name 1]	[Roll Number 1]
[Student Name 2]	[Roll Number 2]
[Student Name 3]	[Roll Number 3]
[Student Name 4] *(if any)*	[Roll Number 4] *(if any)*

Statement of Submission

This is to certify that the following students have successfully completed the final year project titled **"[Title of the Final Project]"** at **[College Name]**, in partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science**:

Name of Student	Roll Number
[Student Name 1]	[Roll Number 1]
[Student Name 2]	[Roll Number 2]
[Student Name 3]	[Roll Number 3]
[Student Name 4] *(if any)*	[Roll Number 4] *(if any)*

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Project Supervisor

[Name of the Project Supervisor]

[Designation]

[College Name]

Acknowledgement

We sincerely acknowledge the invaluable cooperation and support provided by **[Name of Acknowledger]**, **[Designation]**, **[Address of Organization]**. Their continuous guidance has been a great source of motivation throughout the course of this project.

We would also like to extend our gratitude to **[Name of Second Acknowledger]**, **[Designation]**, **[Address of Organization]** for their helpful insights and encouragement during the development of our project.

Our heartfelt thanks go to our friends and families, whose silent support and encouragement played a vital role in helping us complete this project successfully.

Special Thanks:

- Mr. XYZ_1
- Mr. XYZ_2

Date: [Insert Date Here]

Abstract

With the ever-increasing demand for transporting vast amounts of information—such as voice, data, and video—modern telecommunication networks have evolved into all-digital, high-speed broadband systems. Depending on the nature of the information and the network infrastructure, data transmission can occur via **Constant Bit Rate (CBR)** using circuit-switched networks, or **Variable Bit Rate (VBR)** using packet-switched networks. To efficiently utilize network resources and support diverse data types, technologies like **Broadband Integrated Services Digital Network (BISDN)** and **Asynchronous Transfer Mode (ATM)** have been introduced.

This project focuses on the study and application of **Digital Subscriber Line (DSL)** technology—one of the key broadband solutions that enable high-speed transmission of voice, data, and video over existing telephone lines. In addition to exploring DSL fundamentals, this project addresses several critical research challenges in network technology, including **modelling of VBR video traffic**, **traffic class prioritization**, **quality of service (QoS) provisioning**, **optimal bandwidth allocation**, **routing**, and **call admission control**.

The objective is to understand how DSL technology supports QoS and what mechanisms are implemented to ensure reliable and efficient data delivery, especially for multimedia applications. The outcomes of this project contribute to the broader understanding of how broadband networks can be optimized for performance and service quality in modern communication systems.

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Introduction

- Background
- Problem Statement
- Objectives
- Scope of the Project
- Methodology Summary
- Tools and Technologies Used

Literature Review / Related Work

- Summary of similar existing systems or research
- Analysis of gaps or limitations in current solutions
- Justification for your approach

System Analysis

- Requirements Gathering
- Functional Requirements
- Non-functional Requirements
- Use Case Diagrams / Descriptions
- System Models (DFDs, ERDs, etc.*)*

System Design

- System Architecture
- Database Design
- Interface Design
- Component Design
- UML Diagrams (Class Diagrams, Sequence Diagrams, etc.)

Implementation

- Development Tools and Environment
- Modules Implemented
- Screenshots (with brief descriptions)
- Code Snippets (if necessary)

Testing

- Testing Techniques Used
- Test Cases / Test Plans
- Results and Bug Reports

Results and Discussion

- Performance Metrics
- Evaluation of Results
- Comparison with Existing Solutions (if applicable)

Conclusion and Future Work

- Summary of Achievements
- Challenges Faced
- Suggestions for Further Improvement

References

- Follow IEEE / APA / MLA or any other approved citation style

Appendices *(if applicable)*

- Additional Diagrams
- Complete Code Listing
- User Manual / Installation Guide

Typographical Format and Binding

- **Font:** Aptos (entire document)
- **Main Headings:** 16 pt, **bold**
- **Subheadings:** 14 pt, **bold**
- **Body Text:** 12 pt, regular

Page Format

- **Page Size:** A4
- **Top Margin:** 1.75 cm
- **Bottom Margin:** 1.00 cm
- **Left Margin:** 2.00 cm
- **Right Margin:** 2.00 cm

Page Numbering

- Page numbers must appear **at the bottom right**, as part of the **footer**.
- **Title page:** Not numbered
- **Preliminary pages (before Chapter 1):** Use **lowercase Roman numerals** (i, ii, iii, …)
- **Main content (Chapter 1 onward):** Use **Arabic numerals** (1, 2, 3, …)

Header

- Each page must include a header with:
  - **Project Name** (left-aligned)
  - The word **"Documentation"** (right-aligned)
- If the project title is long, use a shortened version for the header.

Footer

- Each page must include a footer with:
  - **"BSCS[Batch] | [Affiliated College Name]"** (left-aligned), where **[Batch]** refers to the batch number (e.g., **F2021**, **S2021**, etc.)
  - **Page numbering** in the format **"Page X of Y"** (right-aligned)
- In case of long project titles, a shortened version may be used in the footer as well.

Chapter Startup

- Each chapter must start on a **new page**.
- The chapter heading should follow this format: **Chapter 1** (cantered both horizontally and vertically) **Chapter Title** (immediately below, also cantered)
- **The actual content** of the chapter must begin **on the next page**.

Text Formatting

- The report must be printed **on both sides of the paper**.
- Content should be aligned appropriately for **double-sided printing**.
- Each page may contain **text, tables, or figures** as required.
- Ensure consistent formatting and readability on **both left and right pages**.

Tables and Figures

- Tables and figures must be placed **on a separate page**, on the **right side** only.
- A single page **may include more than one table or figure**, but **text must not** be interlaced with them.
- Each table/figure must be **numbered** and **captioned** clearly. Examples:
    - Table 1.2: Population distribution in Asia
    - Figure 3.2: Temperature distribution

Binding

- All reports must be **hard bound**, with appropriate **printing on the spine** (backbone).
- **Two hard copies** of the report must be submitted.

Color of Binding: Black

Contents of the Attached CD

Each report/thesis must be submitted with a **CD/DVD** containing the following organized structure:

Doc

This folder should contain all **documentation** related to the project:

- Instructions on how to access the CD and run the project
- All previously submitted reports
- Final project report in thesis format
- Installation instructions
- Troubleshooting guidelines
- User manual
- Research material (including referenced URLs)
- Papers consulted or referred to
- Slides used in project presentations

Source

This folder should include all **source files** required to **compile** the project:

- Subdirectories may be created as needed
- Must include **sample data files** for testing

Project

This folder should contain the **running version of the project**, including:

- Sample data files
- Sample output
- The project must run without errors if copied to another machine
- May include:
    - Executable (.exe) files
    - Installer
    - Or a directly runnable version of the system

Subdirectories may be added with appropriate names for organization.

Length

The length of your dissertation depends on the nature and scope of the project you have undertaken. An excellent dissertation is often **concise yet comprehensive**, effectively conveying key ideas without unnecessary length.

Lengthy datasets, logs, code, or supplementary material should be submitted electronically on the attached **CD/DVD**, rather than included in the printed document.

Binding Side (Spine Text)

The **spine of the bound report** must display the following information **vertically**, in the exact order and format shown below:

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[Title of the Project] BSCS Fall [####].